

SEBASTIAN TUDOR GHERGHE

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EDUCATION

PhD in Mathematics <i>Department of Mathematics, University of Toronto. 3.96/4.00.</i> - Thesis: Adiabatic quantum molecular dynamics, sup. Prof. Israel Michael Sigal. - Awards: QEII/Halperin OGS & Ida Bulat Teaching Awards (2023), Blyth & UofT Top Up Fellowships (2022).	2019–2024 <i>Toronto.</i>
MSc in Mathematics <i>Department of Mathematics, University of Toronto.</i> - Mathematical analysis of Schrödinger–Poisson systems, sup. Prof. I. M. Sigal. - Awards: Margaret Isobel Elliott (2019).	2018–2019 <i>Toronto.</i>
B.Sc. (Hons) Specialist in Mathematics and Physics <i>Faculty of Arts and Science, University of Toronto.</i> Awards: NSERC USRA (2017), George Roderick Fraser Scholar (2015–2017) & Admission (2014).	2014–2018 <i>Toronto.</i>

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, Matplotlib), L^AT_EX, Git.

Data & ML: Data wrangling, supervised learning (linear regression, binary classification), unsupervised learning (PCA, clustering), basic time-series analysis.

Math: PDEs, spectral theory, many-body quantum mechanics.

Languages: English (native), Romanian (native), Spanish (basic).

NOTABLE EXPERIENCE

Data Science Bootcamp <i>The Erdős Institute. Certificate.</i> Collaborated on a final project to scrape 3 000 World Bank loan pdfs for regional/textual analyses. (Python: pandas, NumPy, scikit-learn, NLTK, Matplotlib; PCA).	2021 <i>Online.</i>
Course Instructor <i>Department of Mathematics, University of Toronto.</i> Lectured 7 undergraduate courses (150 students). Designed syllabi/exams and supervised up to 3 TAs.	2023–2024 <i>Toronto.</i>
Academic Research/Publications <i>University of Toronto.</i> - On the Time-Dependent Born-Oppenheimer Approximation (2024), <i>S. Gherghe, I. Moyano, and I. M. Sigal</i>. Submitted; preprint. - On the Mixed Quantum-Classical Approximation (2024), <i>S. Gherghe and I. M. Sigal</i>. Preprint. - Global Spectral Gap in Bosonic Molecular Hamiltonians (2023), <i>S. Gherghe</i>. Preprint. - Manipulating femtosecond laser interactions in bulk glass and thin-film with spatial light modulation (Conference presentation, 2017), <i>E. Alimohammadian, S. Ho, E. Ertorer, S. Gherghe, J. Li, and P. R. Herman</i>. Proc. SPIE 10094, Frontiers in Ultrafast Optics: Biomedical, Scientific, and Industrial Applications XVII, 100940Z. DOI.	2016–2024 <i>Toronto.</i>
Undergraduate Research Assistant <i>University of Toronto.</i> NSERC USRA to study stability of matter/ovals conjecture in mathematics (2017); paid lab work in high-powered ultrafast laser materials processing (2016), cyclic voltammetry (2015).	Summers 2015–2017 <i>Toronto.</i>

PROJECTS

Olliver-Ricci curvature as a fragility indicator <i>Department of Mathematics, University of Toronto.</i> - Co-authored a preprint investigating an optimal transport graph-theoretic stock market fragility indicator. Applied it to S&P 500, NDXT, and TMX data; showed it quantifies crisis depth and duration (e.g. Lehman 2008). - Tools: Python (pandas, NumPy, yfinance), optimal transport, graph theory, GitHub.	2024 <i>Toronto.</i>
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